# Prodigy Anesthesia Quick Review

## Intracranial Surgery

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| List various treatments that will lower intracranial hypertension and cerebral edema. | Fluid restriction, mannitol, furosemide, corticosteroids, and hyperventilation. These actions lower intracranial hypertension and cerebral edema, which in turn lower the ICP.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 596.* |
| Why is it necessary to hyperventilate a patient during surgery for an intracranial aneurysm? | Hyperventilation decreases PaCO2, which decreases the cerebral blood volume, and provides brain relaxation. The goal is to maintain PCO2 levels between 30 and 35mmHg.<br>*Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 8.* |
| Upon induction of anesthesia, what are three things that can be deleterious to the neurosurgical patient? | Apnea, hypertension, and hypotension.<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1009.* |
| How is mild hypothermia helpful in craniotomies for intracranial aneurysms? | It causes a reduction in CMRO2 and reduces the chance of ischemic injury while the surgical clip is in place. There is an approximate 30% reduction in CMRO2 at 33 degrees Celsius. At temperatures between 33 and 34 degrees C, there is minimal incidence of cardiac dysrhythmias, and there is little effect on coagulation. This can be attained through bladder irrigation and water-circulating pads.<br>*Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 9.* |
| What is meant by the term Triple H therapy? | It refers to Hypervolemia, Hypertension, and Hemodilution. This term is in reference to patients with symptomatic vasospasms.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 605.* |
| What is the desired level of anesthetic if brain relaxation is desired and SSEP monitoring is used during a neurosurgical case? | Less than 1 MAC of volatile anesthetic. Opioids are often employed to reduce the anesthetic requirement of the volatile agent.<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1010.* |
| What is the opioid of choice during neurosurgery if extubation is desired at the end of the case, and a neurological assessment is desired? | Remifentanil<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1010.* |

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| Hyperventilation is one method used in providing brain relaxation during neurosurgery. Name 3 other substances that produce brain relaxation. | Hypertonic Saline, IV anesthetics, and mannitol<br><br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1010.* |
| What drugs are appropriate for controlling hypertension when emerging neurosurgical patients? | Apresoline, nicardipine, and labetalol<br><br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1012.* |
| What actions would be appropriate to minimize coughing upon emergence from neurosurgery? | 1-1.5mg/kg IV lidocaine, 4% lidocaine put directly into the ETT<br><br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1012.* |
| At what age does bleeding from an AVM usually present? | between the ages of 10 and 30<br><br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 607.* |
| What is the most common symptom of an unruptured aneurysm? What is the most common sign? | Headache is the most common symptom, while a oculomotor nerve palsy is the most common sign.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 604.* |
| What is the incidence of cerebral vasospasm in patients that have undergone surgery to repair a ruptured aneurysm? When is it most likely to occur? | 30% of patients experience cerebral vasospasm that usually occurs between 4 to 14 days after surgery for ruptured aneurysm repair.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 642.* |
| What are the anesthetic goals for the patient undergoing surgery for intacranial aneurysm? | Adequate CPP, reduction of CMRO2, and to provide an optimal working space within the surgical site by decreasing intracranial volume (tissue and blood), which lessens the requirement of surgical retraction of the brain tissue.<br>*Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 8.* |
| Why should patients undergoing craniotomies in the sitting position be evaluated for a patent foramen ovale? | Patent foramen ovales increase the risk for paradoxical embolisms.<br><br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1012.* |

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| What electrolyte disorder can precipitate or worsen cerebral edema? | Hyponatremia causes the extracellular fluid to become hypotonic. As this occurs, water begins moving into the brain cells causing cerebral edema and intracranial hypertension.<br>*Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 359.* |
| In what patient populations should mannitol be avoided? | Mannitol is an osmotic diuretic that transiently increases intravascular volume, therefore it should not be used in patients with borderline renal or cardiac function. Caution should also be used when administering mannitol in patients with AVMs and intracranial aneurysms. Mannitol can expand hematomas, therefore its use should be limited to the open cranium.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 594.* |
| For patients presenting with intracranial hypertension, why is it best to avoid premedication? | Premedication results in respiratory depression, which in turn causes hypercapnia. Increased PCO2 results in increased ICP.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 633.* |
| Blood flow to the brain is provided by what arteries? | 2 vertebral arteries, 2 carotid arteries, and collateral circulation is provided by the Circle of Willis<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 996-997.* |
| A patient is undergoing a craniotomy in the sitting position. A venous air embolism is detected. What actions do you take. | The surgeon must first be notified in order for the surgical field to be flooded and packed until the site of entry can be detected. The patient should be given 100% O2, and aspiration of the central venous catheter should be done to retrieve entrained air. The CVP should be increased through intravascular volume infusion, and any hypotension should be treated with vasopressors. Also helpful to the surgeon in indentifying the location of the embolism, is compression of both jugular veins. This increases CVP, slowing down air entrainment and produces back bleeding. The patient should be placed head down and the wound closed as quickly as possible if the above measures are not effective.<br>*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 600.* |
| CO2 greatly affects CBF. How does O2 affect CBF? | Little effect on CBF is seen until excessively low PaO2 levels are reached. CBF begins to increase greatly when the PaO2 falls to less than 50 mm Hg.<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 999.* |

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| How does Ketamine affect cerebral blood flow and metabolism? | Ketamine increases both cerebral blood flow and metabolism.<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 999.* |
| The IV agents thiopental and propofol exert what effects on CMRO2 and CBF? | Thiopental and propofol reduce metabolism and cause cerebral vasoconstriction (lower CBF).<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 999.* |
| How do inhalational agents affect CBF? | Inhalational agents cause vasodilation in a dose-related fashion. At low doses, little to no change is seen, however with higher doses, CBF is increased.<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 999.* |
| A patient has developed the Cushing reflex (also called the Cushing triad). Describe this phenomena. | The Cushing reflex consists of bradycardia, hypertension, and respiratory irregularity. It is seen when ICP levels rise so significantly, that brain stem herniation occurs.<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 484.* |
| Succinylcholine should be avoided in what neurosurgical populations? | Patients with muscle denervation from myelopathy, spinal cord injury, or stroke. (Succinycholine is acceptable during the acute phase of stroke and spinal cord injury if within 24-72 hours of the injury). These patient populations have an up-regulation of acetylcholine receptors resulting in severe hyperkalemia upon the administration of succinylcholine.<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1010.* |
| List four interventions for inadequate cerebral perfusion pressure (CPP). | 1. Decrease cerebral blood volume 2. Remove CSF 3. Reduce brain water 4. Elevate MAP<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1019.* |
| What are some interventions that elevate MAP, thus enhancing CPP? | Administration of vasopressors, intravascular volume resuscitation<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1017.* |
| List some mechanisms used to remove CSF. | The intravenous administration of hypertonic fluid and the placement of lumbar or external ventricular drains<br>*Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1019.* |

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